PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

GreenLedger: Development of a Role-Based Secure Records and Payment Management System for Memorial Park Services

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Organizations that manage client services and financial transactions require reliable systems to store and monitor important records. Proper record management is essential to ensure that client information, service details, and payment transactions are accurately documented and easily accessible when needed. In service-oriented industries, maintaining organized records helps improve operational efficiency and supports better decision-making.

In the memorial park service industry, organizations handle various records such as burial lot ownership, client information, and payment transactions. These records must be maintained accurately to ensure proper management of services and financial accountability. However, many organizations in this sector still rely on manual documentation methods such as paper-based records or basic spreadsheets for managing their operational data.

Manual record management often leads to several operational inefficiencies. Physical records can be misplaced, damaged, or difficult to retrieve when needed. In addition, tracking payment transactions manually may result in delays in updating records and increase the risk of human errors. Another concern is the lack of proper control over access to sensitive information, which may allow unauthorized users to view or modify important records.

To address these challenges, digital management systems can be used to improve record organization, transaction monitoring, and information accessibility. One important feature of modern systems is Role-Based Access Control (RBAC), which allows administrators to assign user permissions according to their roles and responsibilities. Therefore, this study proposes the development of **GreenLedger: Development of a Role-Based Secure Records and Payment Management System for Memorial Park Services**, which aims to provide a secure and organized platform for managing client records and payment transactions while ensuring controlled access to sensitive information.

1.2 Statement of the Problem

Despite the reliance of many organizations on digital tools, Greengarden Inc. still faces challenges in managing client records, burial lots, and payment transactions. This study seeks to address the following problems:

1. The current system is manual or spreadsheet-based, making record management time-consuming and prone to errors.
2. Payment transactions are not tracked efficiently, leading to delays and potential financial mismanagement.
3. Sensitive client and financial information is not fully protected, risking unauthorized access or modification.
4. User actions and modifications are not logged, making it difficult to monitor and audit activities.
5. Backup systems are insufficient, increasing the risk of data loss in case of hardware failure or cyber incidents.

1.3 Objectives of the Study

1.3.1 General Objectives

The general objective of this study is to design and develop GreenLedger, a secure, role-based records and payment management system for Greengarden Inc., ensuring accurate record-keeping, controlled access, and efficient transaction monitoring.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze the current manual and semi-digital systems of Greengarden Inc. and identify limitations.
2. Design a secure and user-friendly login and registration interface with role-based access.
3. Develop modules for client record and burial lot management.
4. Implement a payment tracking system to monitor financial transactions accurately and in real-time for authorized personnel.
5. Integrate activity logs to track and audit user actions.
6. Implement office-restricted access and backup systems to ensure data security and recovery.
7. Test and evaluate the system’s security, usability, and performance.

1.4 Scope and Delimitation

The study focuses on the development of GreenLedger, a web-based, role-based records and payment management system for Greengarden Inc.. The system will include user login and registration with role-based access for administrators, accounting staff, and general staff, client record and burial lot management, payment tracking for authorized personnel, activity logs for auditing user actions, office-restricted access, and backup systems to ensure data security and recovery. The system will be designed to operate within the organization's office network and incorporate cybersecurity measures to protect sensitive client and financial information.

The study will not cover public access or open registration for clients, integration with external payment gateways, deployment across multiple branches, or large-scale database operations. The system is developed as a prototype for academic purposes and internal use within Greengarden Inc. only.

1.5 Significance of the Study

The proposed system will benefit the following:

- **Administrators** – By enabling effective supervision, monitoring, and auditing of records and financial transactions.
- **Accounting / Financial Staff** – By providing accurate, real-time payment tracking and reliable reporting.
- **General Staff / End Users** – By simplifying client record management while restricting access to sensitive financial data.
- **Clients** – By ensuring their personal and payment information is securely managed.
- **Greengarden Inc**. – By improving operational efficiency, reducing errors, and enhancing data security.
- **Cybersecurity Students / Researchers** – By serving as a practical example of implementing secure, role-based systems.
- **Future Developers** – By providing a model for secure application design with login, RBAC, auditing, and backups.

1.6 Definition of Terms

- **Activity Logs** – Records of user actions in the system to ensure accountability and traceability.
- **Backup System** – A method of copying and storing data to prevent loss in case of system failure or cyber incidents.
- **Greengarden Inc.** – The memorial park organization that manages burial lots, client records, and financial transactions.
- **Login / Authentication** – The process of verifying a user’s identity before granting access to the system.
- **Memorial Park Services** – Services offered by Greengarden Inc., including burial lot management and interment services.
- **Payment Tracking** – Monitoring and recording client payments accurately, accessible only to authorized personnel.
- **Role-Based Access Control (RBAC)** – A method of restricting system access based on a user’s role in the organization.
- **Secure System** – A system designed to protect sensitive data from unauthorized access or modification.
- **Office-Restricted Access** – Limiting system use to authorized personnel within designated office locations or networks.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Sandhu, Coyne, Feinstein, and Youman (1996) introduced the concept of Role-Based Access Control (RBAC) as a method for restricting system access based on user roles within an organization. Their study explained that RBAC assigns permissions to roles rather than individual users, making it easier to manage system security and access privileges in multi-user environments. This approach improves accountability and prevents unauthorized users from accessing sensitive system information. Similarly, Ferraiolo, Kuhn, and Chandramouli (2003) emphasized that access control mechanisms play a critical role in protecting digital information systems. Their research highlighted that RBAC helps organizations enforce security policies by ensuring that only authorized users can perform specific system functions. These studies are relevant to the present study since the proposed system implements role-based access to control administrative functions and protect sensitive records within the system.

Sandhu, R., Coyne, E., Feinstein, H., & Youman, C. (1996). Role-based access control models. *IEEE Computer*, 29(2), 38–47. https://doi.org/10.1109/2.485845

Ferraiolo, D., Kuhn, D., & Chandramouli, R. (2003). *Role-Based Access Control*. Artech House. https://csrc.nist.gov/projects/role-based-access-control

2.2 Local Studies

Gamba, Guevara, and Naval (2014) examined the implementation of electronic records management systems in Philippine government institutions and found that digital systems significantly improve the storage, organization, and retrieval of administrative records compared to traditional manual processes. Their study emphasized that electronic records systems reduce the risk of document loss and improve operational efficiency in managing institutional data. Similarly, Anyayahan (2017) discussed the importance of digital records management practices in Philippine organizations and concluded that computerized systems improve data accessibility and reduce human errors commonly associated with manual record keeping. These findings highlight the importance of implementing digital systems in organizations, which supports the objective of the present study to develop a web-based administrative system that manages records efficiently.

Gamba, M., Guevara, R., & Naval, V. (2014). Electronic records management in the Philippine government: Issues and challenges. *Philippine Journal of Librarianship*, 35(2), 45–60. https://ejournals.ph/article.php?id=7961

Anyayahan, M. (2017). Digital records management practices in Philippine institutions. *Philippine Journal of Information Studies*, 12(1), 15–28. https://ejournals.ph/article.php?id=11934

2.3 Related Literature

Pressman and Maxim (2019) emphasized that structured software development processes help ensure the reliability and quality of information systems. In modern system development, Agile methodology has become widely used because it promotes flexibility, collaboration, and continuous improvement during the development process. According to Beck et al. (2001), Agile development focuses on iterative progress, allowing developers to continuously refine system features based on feedback and changing requirements. Additionally, Behl and Behl (2017) highlighted the importance of system monitoring and audit logs in maintaining system security and accountability. Audit logs allow administrators to track user activities within the system, detect unauthorized actions, and maintain a record of system transactions. These concepts support the present study since the proposed system uses Agile methodology during development and includes activity logs to monitor user actions and improve system security.

Pressman, R. S., & Maxim, B. R. (2019). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill Education.
Beck, K., Beedle, M., Van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M.,(2001). Manifesto for Agile Software Development. Agile Alliance. https://agilemanifesto.org
Behl, A., & Behl, S. (2017). System monitoring and audit logs for cybersecurity. *International Journal of Computer Applications*, 160(7), 15–22. https://www.ijcaonline.org/archives/volume160/number7/behl-2017-ijca-914749.pdf

2.4 Synthesis of Reviewed Literature

The reviewed foreign and local studies emphasize the importance of secure and efficient information systems in managing organizational data. Foreign studies highlighted the role of

Role-Based Access Control (RBAC) in protecting system resources and ensuring that only authorized users can access sensitive information. Local studies demonstrated that digital records management systems improve the storage, organization, and retrieval of records while reducing errors associated with manual processes. Related literature further emphasized the importance of Agile methodology in developing reliable systems and the use of audit logs for monitoring user activities and maintaining system accountability. However, limited studies focus on integrated systems specifically designed for memorial park services that combine secure record management, payment tracking, and role-based access control in a single platform. Therefore, this study develops **GreenLedger**, a role-based records and payment management system for **Greengarden Inc.** to improve record organization, transaction monitoring, and controlled access to sensitive information.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a **developmental research design** to develop **GreenLedger: A Role-Based Secure Records and Payment Management System for Greengarden Inc.** This design focuses on the creation, design, and evaluation of a web-based system intended to improve the management of client records, burial lot information, and payment transactions within the organization.

The system will be developed based on the identified problems in the existing manual or spreadsheet-based record management process. The proposed system aims to improve data organization, transaction monitoring, and access control to ensure efficiency, accuracy, and security in handling sensitive client and financial information.

3.2 System Development Methodology

This study will use the **Agile methodology** in developing the **GreenLedger Records and Payment Management System**, consisting of the following phases:

Requirements Analysis
Information about the current record management process of Greengarden Inc. will be gathered through interviews, observation, and document review. This phase identifies the system requirements and security needs of the organization.

System Design
The system structure and workflow will be designed using diagrams such as Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), and Use Case Diagrams. These diagrams illustrate how data moves within the system and how different components interact.

Development
The system will be implemented using **Python with the Django framework** for the backend and **HTML, CSS, and JavaScript** for the frontend interface.

Testing
The developed system will undergo several tests to ensure that each module functions correctly and that the system performs according to the specified requirements.

Deployment (Prototype)
The system prototype will be deployed within the organization’s office environment for demonstration and evaluation purposes.

Continuous Improvement
Feedback gathered from users and system evaluation will be used to improve the system’s functionality, usability, and security.

3.3 System Architecture

The **GreenLedger Records and Payment Management System** follows a client-server architecture with a three-tier structure: **presentation layer, application layer, and data layer**. This separation ensures efficient communication between the user interface, system processing, and data storage.

Users, including administrators, accounting staff, and general staff, access the system via a web-based interface. The **presentation layer**, built with HTML, CSS, and JavaScript, allows users to log in, manage client records, and record payment transactions.

The **application layer**, developed in Python using Django, handles core system functions such as authentication, role-based access control (RBAC), payment tracking, and activity logging. Django also provides security features like password hashing and input validation to protect user data.

The **data layer** uses an SQLite database to securely store client records, payment transactions, user accounts, and activity logs. This setup enables fast retrieval and updates of information whenever requested.

Overall, this architecture ensures organized communication between system components while maintaining secure access and efficient management of sensitive data for **Greengarden Inc.**

3.4 Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Django Framework)

- Database: SQLite
- Development tools: Visual Studio Code

3.5 Data Gathering Techniques

- **Interviews** – Conducted with the manager of GreenGarden Inc. to understand the current workflow, challenges, and requirements for client records and financial accounting. This provided insights into what features are needed for the proposed system.
- **Document Analysis** – Reviewed existing ledgers, client records, and memorial lot plans provided by the manager. This helped determine the necessary data fields and informed the design of a secure and

3.6 Testing Procedures

- Unit Testing
- System Testing
- User Acceptance Testing

3.7 Evaluation Criteria

The system will be evaluated based on:

- Functionality
- Usability
- Security
- Efficiency
- Reliability

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

User → GreenLedger System → Database

Process Explanation:

- User logs in and submits data (records or payments)
- System processes and validates the input
- Data is stored in the database
- System retrieves and displays requested information

Entity Relationship Diagram (ERD)

Entities:

- **ClientPersonalInfo** (id, client_first_name, client_middle_name, client_last_name, client_address, client_contact_number, client_civil_status, client_date_birth, client_religion, client_occupation, client_employer_name, client_implyer_address, client_spouse_name, client_spouse_date_birth, client_spouse_occupation, client_spouse_employer, client_id_type, client_id_number, client_date_issued, client_place_issued

- **Beneficiary (**client, name, relationship)

- **ClientStatus** (id, client, plan, monthly_payment, duration, months_remaining, start_date, balance, paid_balance, date_paid, status)

- **Payment** (client_status, month, amount, is_paid, date_paid, ordering, status)

- UserLog (id, role, first_name, middle_name, last_name, date_of_birth, government_id, phone_number, address, email, emergency_contact_name, emergency_contact_number, time_in, time_out, activities, pin)

Relationships:

- User manages Client records
- Client makes Payment
- User generates Activity Logs

Use Case Diagram

Actors:

- Administrator
- Accounting Staff
- General Staff

Use Cases:

- Login / Logout
- Manage Users (Admin)
- Manage Client Records
- Record Payments (Accounting)
- View Reports
- View Activity Logs (Admin)